

HIGH INTENSITY TRAINING (HIT) INFOGRAPHICS COLLECTION ***FEBRUARY 2025 UPDATE***

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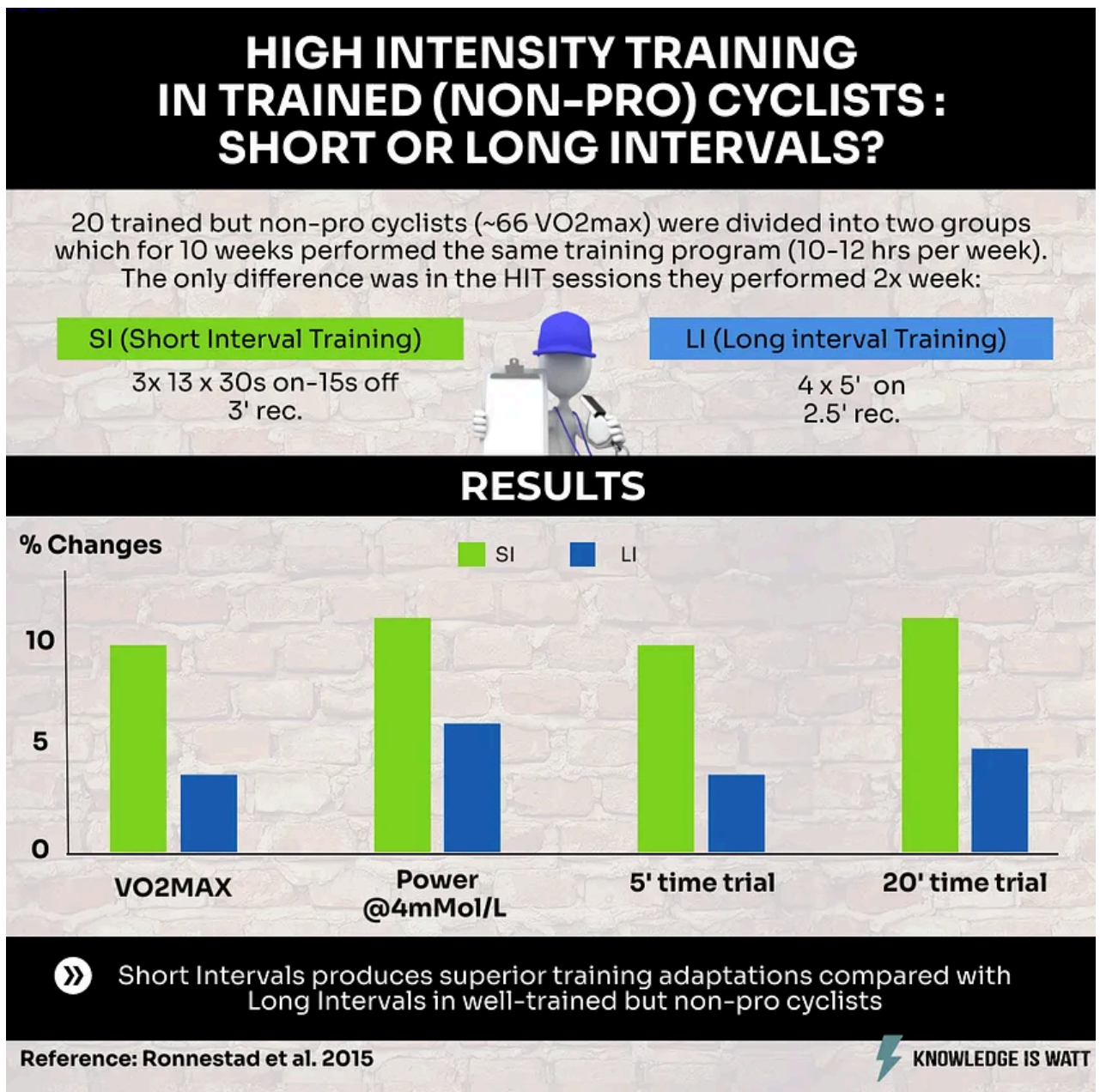
Knowledgeiswatt

February 10, 2025

Here is the growing infographics library about all the relevant researches on **high intensity training and cycling performance** with links to each study's summary.

1. SHORT VS LONG HIT INTERVALS IN WELL-TRAINED (NON-PRO) CYCLISTS:

[Research Summary and Comment](#)



2. SHORT VS LONG HIT INTERVALS IN PRO CYCLISTS

[Research Summary and Comment](#)

SHORT VS LONG HIGH INTENSITY INTERVALS IN PRO CYCLISTS

18 elite cyclists were divided into two groups and performed 3 times per week for 3 weeks two different interval training sessions performed at the maximum sustainable intensity (max effort).

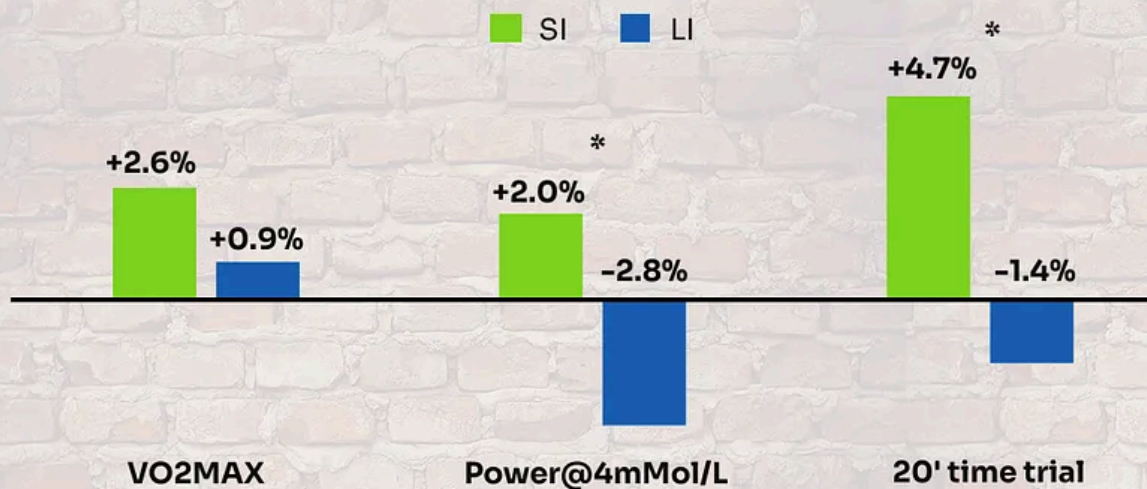
SI (Short Interval Training)

3x 13 x 30s on-15s off
3' rec.

LI (Long interval Training)

4 x 5' on
2.5' rec.

RESULTS



Short high intensity intervals produce superior training adaptations compared to Long Intervals in professional cyclists

Reference: Ronnestad et al. 2020



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3. LONG HIT INTERVALS: FAST START VS STEADY PACE

[Research Summary and Comment](#)

FAST-START LONG HIT INTERVALS

11 well-trained cross-country skiers ($\text{VO}_{2\text{max}}$ 70) completed 2 different HIT (High Intensity Training) sessions. MAP is Maximal Aerobic Power.

FAST START LONG INTERVALS

- 5x5min as follows:
- The first 1min30s at 100% MAP
 - Then, 3min30s at 85% MAP
3min rec.



STEADY PACE LONG INTERVALS

5x5min at 90% of MAP with
3min recovery.



RESULTS

Time
> 90%
 $\text{VO}_{2\text{MAX}}$



RPE



In fast start long intervals, a higher fractions of $\text{VO}_{2\text{max}}$ is sustained, despite a lower RPE (perception of effort).
This could help to maximize performance adaptations

Rønnestad et al. 2020



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4. HIT LONG INTERVALS: 4X8' BETTER THAN 4X4' OR 4X16'

[Research Summary and Comment](#)

HIT LONG INTERVALS: 4x8' BETTER THAN 4x4' OR 4x16'

For 7 weeks, 35 well-trained amateur cyclists (age ~40, VO2max ~52) were divided into 4 different training groups:

* all the following HIT sessions were performed at the maximum sustainable intensity

CONTROL (CON)

they just rode about 9 hours per week at low intensity'.

4X4MIN

they completed 4hrs at low intensity + 2xweek 4x4 min with 2min recovery HIT sessions.

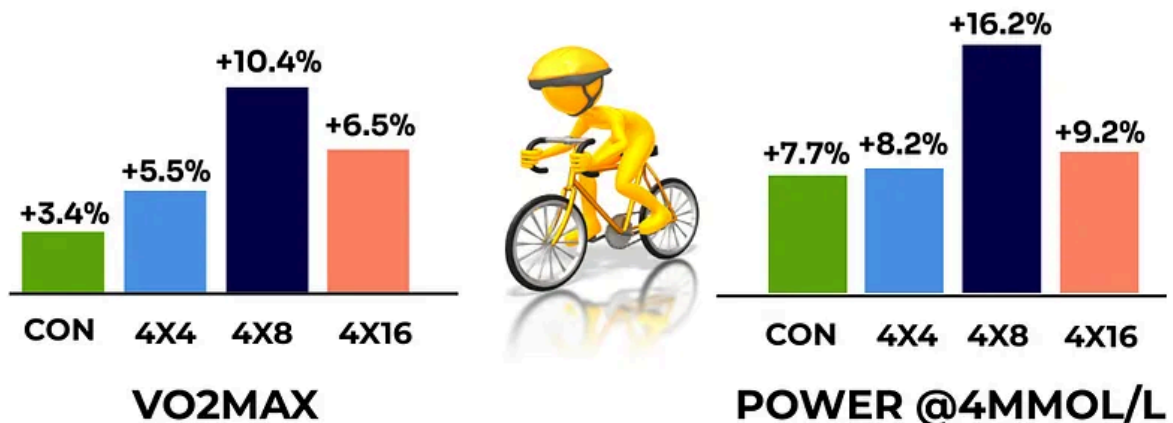
4X8MIN

they completed 4hrs at low intensity + 2xweek 4x8 min with 2min recovery HIT sessions.

4X16MIN

they completed 4hrs at low intensity + 2xweek 4x16 min with 2min recovery HIT sessions.

% CHANGES PRE-POST THE 7 WEEKS



When considering long high intensity intervals performed at the maximum sustainable intensity, 4x8 min seems a better solution compared to 4x4min and 4x16min to induce superior aerobic physiological and performance adaptations.

Reference: Seiler et al. 2013



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5. THE KEY TO DESIGN THE AEROBIC HIT WORKOUTS!

[Research Summary and Comment](#)

THE KEY TO DESIGN THE AEROBIC HIGH INTENSITY TRAINING WORKOUTS!



A very recent study by Odden et al. (2024) collecting data during many HIT sessions, reported that: sustaining a higher fraction of maximal oxygen uptake ($\text{VO}_{2\text{max}}$) during HIT sessions leads to superior improvements in aerobic physiological and performance.



How can be used this information?



- » Performing HIT sessions that permits to sustain a high fraction of $\text{VO}_{2\text{max}}$ for the longer time possible.
- » Which HIT session? Looking at this and other study that recorded oxygen consumption during HIT efforts we can identify some HIT structure that maximise this concept.

Ref. Odden et al. 2024



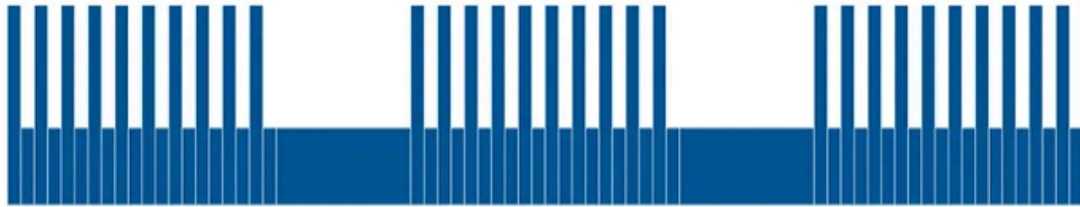
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1. Intermittent Short Intervals

Format:

3 x 8-13 min (with 3-5 min recovery between series) alternating 15-40s 'on' and 15-30s 'off'.



Intensity:

- on phases: 120-130% FTP, while
- off phases: Z1/Z2 in a 5/7 zones model.

Working on this format, you can build a progression through the weeks starting from a on-off ratio 1:1 and progressing to a 2:1, while also increasing the duration of each series.



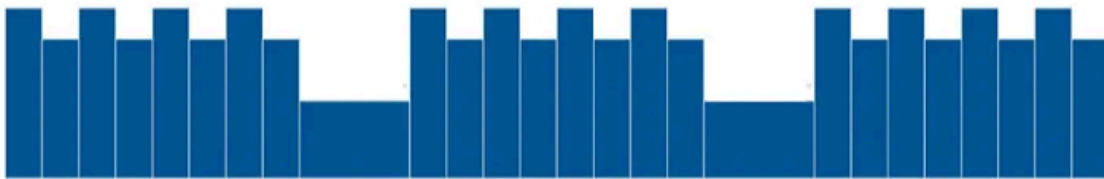
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2. Under-Over Intervals



Format:

3-5 (series) x 8min with
alternating 60s just above FTP/second
threshold/CP (105-110%) and 60s just below
the same intensity (90% FTP).
3-5 min recovery between series.



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3. 8 min constant pace intervals

Format:

3-5 x 8 min @105/110% of FTP with 3-8min recovery.



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4. 5 min fast start intervals



Format:

4-6 x 5 min alternating 1m30s @120% FTP
and 3min30s at FTP. 3-5 min recovery
between series.



6. OPTIMIZING HIT AEROBIC ADAPTATIONS BASED ON RIDER PROFILE

[Research Summary and Comment](#)

ON AVERAGE, WHICH FORMAT IS THE BEST TO OPTIMIZE AEROBIC ADAPTATIONS?

19 trained amateur cyclists (~65 VO₂max) performed on 3 different days, 3 different high intensity training sessions in the lab. In all the 3 formats, the recovery between the 5x8min efforts was 3min at 35% of PO₄₀min. PO₄₀min is the personal best mean power output for 40min.

INTERMITTENT SHORT INTERVALS (ISI)

5x8min alternating 30-s at 118% of PO₄₀min and 15-s at 60% of PO₄₀min

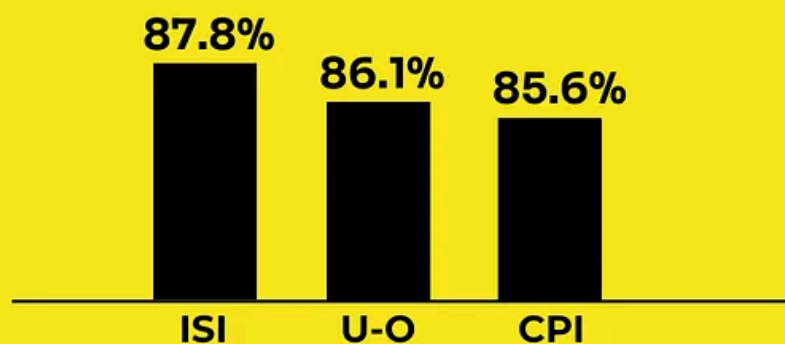
UNDER-OVER INTERVALS (U-O)

5x8min alternating 60-s at 110% and 90% of PO₄₀mi

8' CONSTANT PACE INTERVALS (CPI)

5x8min constant at 100% of PO₄₀min

AVERAGE OXYGEN CONSUMPTION (AS % OF VO₂MAX) ATTAINED DURING THE DIFFERENT FORMATS



On average, intermittent short intervals are the HIT format that permits to accumulate more time at high fraction of VO₂max and so should be the most effective to induce positive aerobic adaptation

Urianstad et al. 2024



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DOES THIS HOLD TRUE FOR ALL THE RIDERS OR IT DEPENDS ON RIDER'S PHYSIOLOGICAL PROFILE?



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THE ANSWER: IT CAN DEPEND ON THE RIDER PHYSIO PROFILE



A recent explorative analysis reported that a higher fraction of $\text{VO}_{2\text{max}}$ during intermittent short intervals is attained in riders with second lactate threshold (LT2) at a relative low % of $\text{VO}_{2\text{max}}$, while in riders with LT2 at high fraction of $\text{VO}_{2\text{max}}$, the under-over and constant pace intervals induce a similar metabolic stress compared to the 30-15 intervals.



How can be used this information?



- » Fostering short intermittent intervals to develop aerobic qualities in riders with second threshold/FTP/CP at a low % of $\text{VO}_{2\text{max}}$. These are probably the punchy and sprinter rider types (?)
- » Riders with second threshold/FTP/CP at a high % of $\text{VO}_{2\text{max}}$ (they should be the climbers and time triallist?) should be able to achieve similar aerobic adaptations by interchangeably using intermittent short intervals, under-overs and 8 min constant pace intervals, with the last two being more specific to their race demands.

Ref. Urianstad et al. 2024



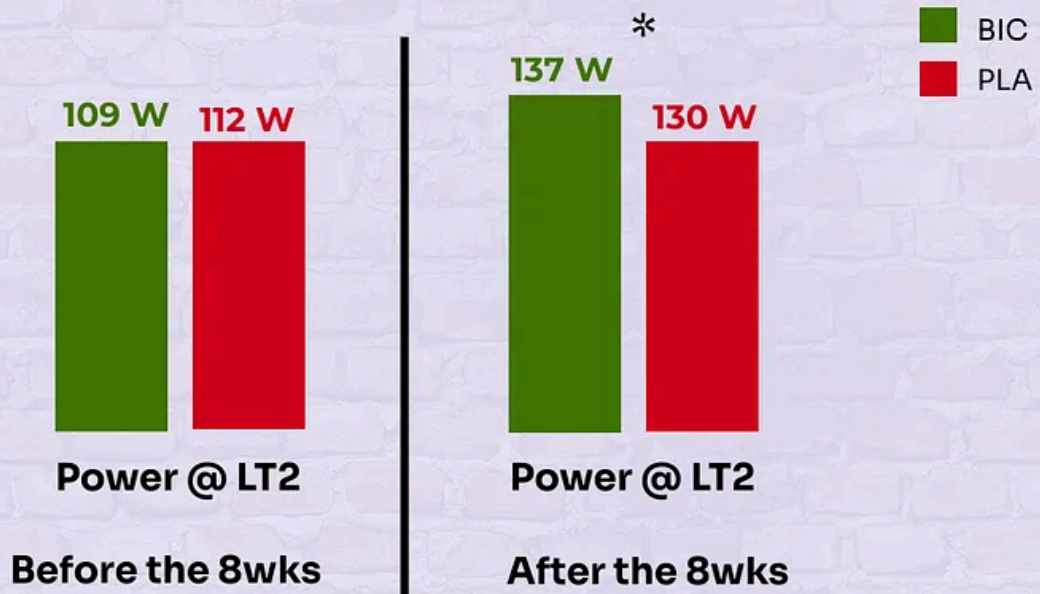
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7. BICARBONATE BEFORE HIT SESSIONS

1. [Research Summary and Comment](#)

BICARBONATE BEFORE HIT TRAINING BOOSTS PHYSIO ADAPTATIONS

16 recreationally active women were placed into the BIC (n = 8) or PLA (n = 8) groups, and performed 8 wk (3 days/wk) of 6-12 X 2min cycling intervals at 140-170% of their second lactate threshold (LT2) (HIT, High Intensity Training). They ingested Bicarbonate (0.4 g/kg) or a placebo before each training session



Bicarbonate supplementation before High Intensity Training can enhance aerobic adaptations



This because reducing metabolic acidosis (H⁺) during HIT could facilitate biological signalling to improve muscle oxidative capacity.

Reference: Edge et al 2006



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LITERATURE BRIEFING:

- Both in well-trained amateur and professional cyclists, short HIT intervals (3x13min alternating 30s on and 15s off) seems to provide superior physiological and performance adaptations compared to Long HIT intervals (4x5min with 2.5min rec.) when both formats are performed at the maximum sustainable intensity. (1, 2)
- When considering long HIT intervals, a fast start for each interval (1m30s at 100% VO₂max power followed by 3m30s at 85% of VO₂max power) permits to spend more time at a high percentage of VO₂max (higher aerobic load) with a lower perceived effort compared to maintain a steady pace (all 5min at 90% of VO₂max power). (3)

- When considering long high intensity intervals performed at the maximum sustainable intensity, 4x8 min seems a better solution compared to 4x4min and 4x16min to induce superior aerobic physiological and performance adaptations. (4)
- A very interesting study in 2024 showed that **sustaining a higher fraction of maximal oxygen uptake ($\dot{V}O_{2\max}$) during HIT sessions leads to superior improvements in aerobic physiological and performance.** (5)

Basing on these principles, here the HIT formats (**in order of effectiveness**) you should pick if you want to optimize HIT's derived aerobic adaptations:

1. **Intermittent short intervals**
 2. **Under-over intervals**
 3. **8 min constant pace intervals**
 4. **5 min fast start intervals**
- Note that the above stated order of effectiveness of these formats is based on studies that evaluated the average responses of a group of cyclists, **without taking into account the physiological profile of each rider.**

A further explorative analysis on this topic (6) showed that:

- **In riders with second lactate threshold at a lower % of $\dot{V}O_{2\max}$ (~ <83%) (sprinters and punchy guys?), the short intermittent intervals permits to spend more time at a high % of $\dot{V}O_{2\max}$, and so they should be more effective than other HIT formats in improving aerobic qualities.**
- **In riders with second lactate threshold at a high % of $\dot{V}O_{2\max}$ (climbers. time trialist?), the under-over and constant pace HIT formats are practically equally effective as the short intermittent intervals in accumulating time at a high fraction of $\dot{V}O_{2\max}$. Given this, the under-over and constant pace HIT could be more appropriate for them, as they represent an effort more similar to their race demands compared to the intermittent short intervals.**
- Ingesting sodium bicarbonate (0.4 g/kg) before HIT sessions for some weeks can boost training adaptations and performance. (7)

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